

Yonatan Tesfagiorgis

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PROFESSIONAL SUMMARY

AI/ML Engineer with experience building retrieval-augmented generation (RAG) systems, NLP pipelines, and deep learning applications using PyTorch, Hugging Face, and vector databases. Skilled in developing end-to-end ML workflows, backend APIs, and scalable AI-powered applications.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML/CSS, PHP

ML/AI: PyTorch, TensorFlow, scikit-learn, Hugging Face, Transformers, NLP, FAISS, LangChain, Codex, Gemini

Backend & Deployment: Flask, FastAPI, React.js, Node.js, Docker, AWS, Azure, Git, CI/CD, REST APIs

PROJECT EXPERIENCE

AI-Powered Document Assistant (RAG + LLMs)

Jan 2026 – Mar 2026

- Designed and deployed an end-to-end RAG pipeline using LangChain, FAISS, Hugging Face, and Streamlit for conversational PDF querying.
- Implemented semantic chunking, vector embeddings, and optimized retrieval workflows to improve response relevance and reduce hallucinations.
- Built low-latency semantic search functionality with FAISS vector indexing for real-time document exploration.
- Deployed an interactive AI chat application on Hugging Face Spaces.

Medical Chest X-Ray Pneumonia Detection

Jan 2026 – Mar 2026

- Developed a deep learning model using PyTorch and DenseNet121 to classify pneumonia and tuberculosis from 15,000+ medical images.
- Applied data augmentation, preprocessing, and class-weighted loss functions to improve model generalization and validation accuracy.
- Evaluated model performance using confusion matrices and classification metrics.

NLP Sentiment Retrieval System

- Built an NLP pipeline for aspect-based sentiment analysis using 10,000+ Amazon product reviews.
- Combined traditional machine learning models with transformer embeddings to improve sentiment classification accuracy.
- Implemented semantic similarity retrieval using cosine similarity and sentence transformers.
- Improved retrieval precision compared to traditional keyword-based sentiment classification methods.

Volunteer Management Application

Jun 2025 – Aug 2025

- Developed a full-stack volunteer management platform using Flask, React, and MySQL with REST API integration.
- Built responsive dashboards, scheduling workflows, and automated PDF/CSV reporting functionality.
- Optimized database queries and backend services for efficient user and event management.

PROFESSIONAL EXPERIENCE

Production Assembly Operator – Randstad

Sep 2020 – Dec 2021

- Improved operational efficiency by 15% through process optimization and workflow coordination.
- Trained team members on safety and quality procedures, contributing to reduced workplace incidents.
- Collaborated with production teams to minimize equipment downtime and improve operational consistency.

LEADERSHIP EXPERIENCE

Cougar CS Organization – University of Houston

Jan 2024 – May 2026

- Led C++ workshops and technical problem-solving sessions for computer science students.
- Mentored peers in programming fundamentals, debugging techniques, and algorithmic thinking.

EDUCATION

University of Houston - B.S. Computer Science

May 2026

Texas State Technical College - A.S. Cybersecurity

Jul 2019